CM52035 – Advanced Computer Vision Coursework 2

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Set	24th Feb, 2026
Due	17th May, 2026, 8 pm
Percentage of overall unit mark	50%
Submission Location	Moodle
Submission Components	Code, log, models, data and report
Submission Format	`.pdf` file named 'StudentID_report.pdf' `.zip` file named 'StudentID_code.zip' `shareable link with other files`
Anonymous Marking	No
Generative AI Assessment Categorisation	Type A (guidance)

*Note: *Due to their large size, models and datasets cannot be directly uploaded to Moodle. Instead, you are required to include their shareable links on the cover page of your report. Please ensure that they are set to be "shared with anyone with the link" and have no changes after the deadline.*

1 Overview

This coursework is for MCom students in the Advanced Computer Vision Unit with Unit code CM52035. You are expected to solve practical problems encountered in real-world computer vision applications. The aim of this coursework is to help you further understand Generative Adversarial Networks (GANs) [1] through the task of image-to-image translation. You will study, implement, and extend either Pix2Pix [2] (paired data) or CycleGAN [3] (unpaired data). Note that you are not required to implement Pix2Pix or CycleGAN from scratch. This coursework will be based on existing code, and you are expected to further modify/extend it.

There are five mandatory tasks and 2 optional ones in this coursework. By completing this coursework, you are expected to:

- Understand the adversarial learning framework behind GANs.
- Explain conditional GANs and cycle-consistency constraints.
- Implement, update and train Pix2Pix or CycleGAN models.
- Evaluate generative models using both qualitative and quantitative metrics.
- Analyse common GAN training issues (mode collapse, instability).

- Propose and justify simple model or training improvements with limited data conditions or various loss function choices.

2 Submission details

This coursework is worth 50 points from the overall mark of 100 for this unit. Marks are specified for each task later in Table 1. A report will be your main method of assessment. Your code will be tested based on the output and performance of your implemented solutions, and cross-checked with your report.

Note that this is an **individual** coursework. You need to submit the project report along with the code, which should include an appropriate README file. Models along with the training logs and datasets are also required to be submitted with a shareable link. It is your responsibility to check that you correctly submit your work. Once you have submitted to Moodle and Online Drive (e.g. OneDrive), you should download your submission and **verify it**.

The submission deadline is **17th May, 2026, 8 pm**. Your submission should be uploaded to Moodle and should consist of the report PDF and the code. Models, training logs and the datasets should be shared as a zip file with a shareable link where we can download *all* the required files. Note that any files that are uploaded or edited after the deadline will **NOT** be assessed.

Reports longer than 10,000 words will not be read. High-quality reports are typically much shorter and more concise. After the main part, please attach the references and appendices, if any (these are excluded from the word count). The submitted code must be well commented (indicating major steps). You should ensure that it is possible for us to run your code and replicate your results. This implies that the code is self-contained without any absolute image paths, and that any README files with environment setup, if required, are provided.

Your report should demonstrate and evaluate your results. It must include figures of appropriate resolutions. Evaluation can be qualitative (how it looks) and, wherever possible, quantitative (tested numerically). You should also concisely explain your approach to solve each task, the choices you make (e.g. hyperparameters, etc). The marks will be decided based on the marking guideline specification that we will follow for this unit.

Late submissions

The university policy will be followed for late submissions. If a piece of work is submitted after the submission date, the maximum possible mark will be 40% of the full mark. If the work is submitted more than five working days after the submission date, the student will receive zero marks. **Requests for extensions should be made to the Director of Studies**. Lecturers and tutors cannot approve extensions. Please make sure you are familiar with the department's coursework deadline extension policy.

3 Tasks

3.1 Task Overview

In this coursework, you will only need to choose **ONE** of the following two tracks:

- TRACK **A**: Pix2Pix (Paired image-to-image Translation). Example: edges $\rightarrow$ photos, semantic labels $\rightarrow$ street scenes, sketches $\rightarrow$ images.
- TRACK **B**: CycleGAN (Unpaired image-to-image Translation). Example: horses $\leftrightarrow$ zebras, summer $\leftrightarrow$ winter scenes, paintings $\leftrightarrow$ photos.

Please confirm the track you choose at the very beginning of your report. You can choose your preferred track, and there will be no difference when we assess your work. For each track, you are expected to (details are provided later):

- Study the related concepts, GAN, Pix2Pix, CycleGAN.
- Study the chosen model and its loss functions.
- Train the model on datasets with different setups.
- Gain a deep understanding of the metrics and evaluate the results.
- Extend and analyse the method with different experiments.

3.2 GAN-related Theory

You are expected to understand the theory behind GAN, Pix2Pix, and CycleGAN. Your report must include a section explaining these techniques, no matter which track you choose. This could include the mathematical explanation along with the details of their neural network architecture design, etc.

3.3 Implementation Requirements

You are NOT required to re-implement Pix2Pix or CycleGAN from scratch. In fact, as is common practice, you should start from a well-established open-source implementation, In this coursework, the official repository must be used as the base implementation for further extension and tests. The official repository is available at: `https://github.com/junyanz/pytorch-CycleGAN-and-Pix2Pix/tree/master`.

Apart from the minimum requirements to run the official code of Pix2Pix or CycleGAN, you can only use the following libraries:

1. PyTorch: torch, torchvision
2. Computing & Data Processing: numpy, scipy, pandas
3. Image Processing: opencv-python (cv2), Pillow (PIL), scikit-image
4. Visualisation & Logging: matplotlib, seaborn, tensorboard, tqdm
5. Metrics related: scikit-learn, torchmetrics. Other metrics implementations are possible only if clearly explained.

The following are strictly **NOT** allowed:

1. Other Generative Model Frameworks: You must not replace the GANs models in this CW with alternative generative models such as diffusion-model-related libraries, pre-trained image-to-image tools, etc.
2. High-Level AutoML Training Tools: This includes AutoML frameworks and automated hyperparameter search platforms such as auto-sklearn, etc.

3. Pretrained Image Translation APIs or Services: External inference services such as HuggingFace hosted pipelines for image translation.
4. Any AI tool to generate entire pieces of code from a text-prompt. This is a class A coursework, so the use of GenAI is not permitted.

If you would like to use other libraries, please check with us before applying them in your coursework.

3.4 Datasets

You can choose one of the datasets provided in the official code.

- TRACK A: Pix2Pix: cityscapes, night2day, edges2handbags, edges2shoes, facades, maps
- TRACK B: CycleGAN: apple2orange, summer2winter_yosemite, horse2zebra, monet2photo, cezanne2photo, ukiyoe2photo, vangogh2photo, maps, cityscapes, facades, iphone2dslr_flower, ae_photos

Alternatively, you can choose your own datasets. Note that your chosen dataset should have at least 400 images in the training set. You are encouraged to use a larger dataset for later extension tasks.

3.5 Extension Tasks

As aforementioned, in this coursework, we are trying to tackle practical problems that you may face in "real-world" scenarios. The following extension tasks (ET) are commonly considered as alternative changes you can make from the given repository for the original methods. Some ETs may be related to the data conditions you will face. Note that you are not expected to obtain "better" performance compared to the original setup. You are required to provide a detailed analysis of these extensions.

3.5.1 ET1: Loss Function Engineering

The provided loss functions in the official repository may work very well on the tested datasets. In real-world scenarios, however, it is very often not the case. Also, new loss functions are often introduced after the tested methods are published. In this first extension task (ET1), you are required to apply at least one novel loss function to the current framework. You can choose to add the new loss function or replace part/all of the original loss function.

For any loss function, you are required to present the motivation for using it, along with the theory behind this loss function. You can use the existing loss function implementation, but you may need to make some changes when applied to your framework.

Try to analyse: With different loss setups and combinations, what results can we obtain? What hyperparameters can we adjust?

3.5.2 ET2: Architecture Design Choices

In this extension task, you are expected to make some changes to the network architecture. Similar to ET1, you need to provide a motivation for this change. You can use the existing implementation and merge it into the current code.

Try to analyse: With different network architectures, what results can we obtain? What hyperparameters can we adjust?

3.5.3 ET3: Evaluation Metrics

In practice, different metrics may be used for different purposes. For example, in this coursework, some may say, "My GAN looks good with well-generated images." However, a qualitative result may not be sufficient, and we need some robust quantitative evaluation pipelines.

In this ET3, you are required to implement at least one additional quantitative evaluation metric beyond those provided in the official repository to evaluate the generated images with different setups.

Try to analyse: With different setups and a new metric, do we get consistent evaluation results? Do we have the setup where the generated images look better, but the quantitative result is worse? Why does this happen?

3.5.4 ET4: Limited Data Conditions

In practice, we often do not have enough training data. In this ET, you are expected to reduce the training data in your datasets and test the robustness of the framework. Please test your framework with the conditions that only 50% and 25% of the training sets are available.

Try to analyse: With a shortage of training data, do we still get consistent evaluation results? If not, how can we improve it?

3.5.5 ET5: Tests based on the mixture of previous ETs

As you have noticed, the previously introduced ETs are design choices where we can test the framework from different perspectives. In this ET, you are expected to have two "setups" that combine previous ETs and do further analysis with quantitative and qualitative results. For example, when we only have 25% of the training data (ET4), are we still keeping the same backbone (ET2)? When we change the backbone(ET2), can we get consistent results using the same evaluation metrics (ET3)?

You are expected to explain these "setups" clearly with details, including the experiments you conducted, hyperparameter tuning, etc.

3.6 Computing choices

In this project, you are suggested to use a machine with GPUs. Here are three recommended ways:

- Use your own laptop/desktop: The tasks in this coursework do not require large GPU resources. Hence, it should be possible to use your own machine to do all the tasks if you have a dedicated GPU (a CPU-only machine will not be enough though).
- Online resources: Online resources such as Google Colab can also be considered. However, if you are using a free version, you may expect some usage limitations.
- Department of Computer Science Server: You can also use the department GPU cluster Hex cloud (`https://hex.cs.bath.ac.uk`). Before you use Hex, please read the wiki (help) pages very carefully and follow the rules when using it. For example, although we have multiple servers on Hex, you can only use some of them based on access rights.

When you submit your code, please mention all the details about the machine you used, e.g. Operating System, GPU version, etc. This will help us replicate and cross-check your results. Note that you cannot use "shortage of computing resources" as a reason to ask for a deadline extension.

4 Criteria for Marking

Task	Mandatory?	Max mark
1. Study and explain the related concepts, GAN, Pix2Pix, CycleGAN	Y	10
2. Train the model on the dataset with default setup and obtain initial results	Y	10
3. ET1: Loss Function Engineering	Y	17
4. ET2: Architecture Design Choices	Y	16
5. ET3: Evaluation Metrics	Y	17
6. ET4: Limited Data Conditions	N	15
7. ET5: Tests based on the mixture of previous ETs	N	15
Max mark with only mandatory tasks		70
Max mark with mandatory tasks and at least one optional task		85
Max mark with all tasks		100

Table 1: Maximum mark for each task.

The maximum mark per task is detailed in Table 1. There is no penalty for not implementing a mandatory task - you will simply get 0 on the tasks that you skip. The marking criteria are explained in Table 2. Note that the report is the most important part of your assessment. We will run your code and check your output.

Please bear in mind that we care more about the explanation of what you have done and the interpretation of the results, not the numbers/results themselves. Below you can find further details about what we expect to see in a good report for each task:

- **Task 1**. The related concepts should be well-explained, incorporating the relevant theory and technical details. You are encouraged to use figures and equations to help illustrate the concepts.
- **Task 2**. Simply explain how you run the code and show the results with the default setup.

Criterion	Description
Explanation of approach	The method is explained well, and it clearly details how the various steps are carried out.
Analysis of performance	The performance is analysed and evaluated well with both quantitative and qualitative evaluation.
Analysis of parameters	The sensitivity of the task with respect to relevant parameters associated with the algorithm are analysed.
Code output	The code runs without errors: can run the training process, testing the given models and replicate the results.

Table 2: Marking criteria.

- **Tasks 3-5 (ET1-ET3)**. What loss function and optimiser did you use? What architecture/metric did you try? If you tried more than one, did they differ in performance? What is the motivation for using these losses/architecture/metric? What are the pros and cons of the original setups? How did hyperparameters affect your results? Analyse common GAN training issues (mode collapse, instability), etc.
- **Task 6 (E4)**. How do you split the datasets? What is the difference compared to the training with the full dataset? Does it get the "best" result faster? How did hyperparameters affect your results? How can you improve the results with limited data? What efforts have you made? Analyse common GAN training issues (mode collapse, instability), etc.
- **Task 7 (E5)**. What design choices did you make for this extension task, and why? When combining these "pre-conditions" in ETs, what findings do you obtain? Do you need a new set of hyperparameters?, etc.

Expected report structure:

1. Introduction
2. Theoretical background
3. Experimental setup/details (hardware info, datasets, training details)
4. Original results
5. Extension tasks with additional results
6. Discussion/Findings
7. Conclusion

5 Feedback and Getting Help

Formative feedback on your work will be offered throughout the duration of the coursework:

- During your labs, the tutors will be available to answer questions and offer guidance. Please note that tutors will not be able to make decisions on your behalf regarding your assignment. They are there to discuss your ideas and offer advice.
- Use Moodle forums to post general questions or questions specific to your assignment. The unit teaching team will respond to these, as well as your peers. This way, we will create a repository of knowledge that will be available to all.

You will receive **summative feedback** on your work within 3 semester weeks of the submission deadline. The feedback will discuss your performance based on the criteria for marking, including what you did well and how specific components/sections could have been improved.

6 Academic Integrity

Your work will be checked to ensure that you have not plagiarised. For more information about the plagiarism policy at the University, see https://library.bath.ac.uk/referencing/plagiarism.

Remember that the published work that you refer to in your report should be clearly referenced in your text and listed in a bibliography section given at the end of your report. For more information, see https://library.bath.ac.uk/referencing/new-to-referencing.

Any code submitted will also be checked.

This is a **Type A** coursework. You cannot use Generative AI (GenAI) tools when completing this assessment. For more information, see this link.

7 FAQs

As this is the first cohort for this coursework, we do not have any questions from the previous students. We will collect your questions and put the FAQs on Moodle later.

References

[1] Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in neural information processing systems*, 27, 2014.

[2] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A Efros. Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1125–1134, 2017.

[3] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE international conference on computer vision*, pages 2223–2232, 2017.